A QUASI-AFFINE SCHEME BECOMES AN AFFINE SPACE AFTER AN AFFINE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the presentation of the quasi-affine scheme as the complement of finitely many equations, the construction of the affine scheme over X from those equations, and the explicit verification of the base change identification by mutually inverse isomorphisms.

Editorial note. The label lemma-quasi-affine-presentation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-quasi-affine-presentation –start 2738 –end 2802 –local-id 28034 returned RESULT=OK (L28034 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of a quasi-affine scheme as a quasi-compact open of an affine scheme. Same chapter, the definition of a quasi-affine scheme and the description of a quasi-compact open of an affine scheme as a complement of a closed subset defined by finitely many equations. Supplies the presentation X = Spec(A) minus V(I) with I generated by finitely many elements f_1, ..., f_n.
- Basic opens form a basis of the spectrum of a ring. Algebra, the section on the spectrum of a ring, where the standard opens D(f) are shown to form a basis of the topology. Supplies the covering of the quasi-compact open by finitely many standard opens, hence the finiteness of the list of equations.

Source reference. Original section 00DY: sources/stacks-chow/tags.txt, source lines 515–515. Prerequisite pointer invoked at proof line(s) 2750; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-affine scheme. There exists an integer $n \geq 0$, an affine scheme T, and a morphism $T \to X$ such that for every morphism $X' \to X$ with X' affine the fibre product $X' \times_X T$ is isomorphic to $\mathbf{A}^n_{X'}$ over X'.*

Date: stacks-project:properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By definition, there exists a ring A such that X is isomorphic to a quasi-compact open subscheme $U \subset \mathrm{Spec}(A)$. Recall that the standard opens $D(f) \subset \mathrm{Spec}(A)$ form a basis for the topology, see Algebra, Section 00DY. Since U is quasi-compact we can choose $f_1, \ldots, f_n \in A$ such that $U = D(f_1) \cup \ldots \cup D(f_n)$. Thus we may assume $X = \mathrm{Spec}(A) \setminus V(I)$ where $I = (f_1, \ldots, f_n)$. We set

$$T = \mathrm{Spec}(A[t, x_1, \ldots, x_n]/(f_1 x_1 + \ldots + f_n x_n - 1))$$

The structure morphism $T \to \mathrm{Spec}(A)$ factors through the open X to give the morphism $T \to X$. If $X' = \mathrm{Spec}(A')$ and the morphism $X' \to X$ corresponds to the ring map $A \to A'$, then the images $f'_1, \ldots, f'_n \in A'$ of $f_1, \ldots, f_n$ generate the unit ideal in A'. Say $1 = f'_1 a'_1 + \ldots + f'_n a'_n$. The base change $X' \times_X T$ is the spectrum of $A'[t, x_1, \ldots, x_n]/(f'_1 x_1 + \ldots + f'_n x_n - 1)$. We claim the A'-algebra homomorphism

$$\varphi : A'[y_1, \ldots, y_n] \longrightarrow A'[t, x_1, \ldots, x_n, x_{n+1}]/(f'_1 x_1 + \ldots + f'_n x_n - 1)$$

sending y_i to $a'_i t + x_i$ is an isomorphism. The claim finishes the proof of the lemma. The inverse of φ is given by the A'-algebra homomorphism

$$\psi : A'[t, x_1, \ldots, x_n, x_{n+1}]/(f'_1 x_1 + \ldots + f'_n x_n - 1) \longrightarrow A'[y_1, \ldots, y_n]$$

sending t to $-1 + f'_1 y_1 + \ldots + f'_n y_n$ and x_i to $y_i + a'_i - a'_i(f'_1 y_1 + \ldots + f'_n y_n)$ for $i = 1, \ldots, n$. This makes sense because $\sum f'_i x_i$ is mapped to

$$\sum f'_i(y_i + a'_i - a'_i(\sum f'_j y_j)) = (\sum f'_i y_i) + 1 - (\sum f'_j y_j) = 1$$

To see the maps are mutually inverse one computes as follows:

$$\begin{gathered} \varphi(\psi(t) = \varphi(-1 + \sum f'_i y_i) = -1 + \sum f'_i(a'_i t + x_i) = t \\ \varphi(\psi(x_i)) = \varphi(y_i + a'_i - a'_i(\sum f'_j y_j)) = a'_i t + x_i + a'_i - a'_i(\sum f'_j a'_j t + f'_j x_j) = x_i \\ \psi(\varphi(y_i)) = \psi(a'_i t + x_i) = a'_i(-1 + \sum f'_j y_j) + y_i + a'_i - a'_i(\sum f'_j y_j) = y_i \end{gathered}$$

This finishes the proof. □